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Influence of ALDH2 Genetic Polymorphisms on Aciclovir Pharmacokinetics following Oral Administration of Valaciclovir in Japanese End-stage Renal Disease Patients

Katsutoshi HARA[1,*], Kazuaki SUYAMA[2], Hiroshi ITOH[3] and Satoru NAGASHIMA[4]

[1]*Clinical Pharmacology Department, GlaxoSmithKline K.K., Tokyo, Japan*
[2]*Clinical Research Department, GlaxoSmithKline K.K., Tokyo, Japan*
[3]*Biomedical Data Sciences Department, GlaxoSmithKline K.K., Tokyo, Japan*
[4]*Shitoro Clinic, Shizuoka, Japan*

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Summary: This study was performed to investigate the pharmacokinetics of valaciclovir (VACV), aciclovir (ACV) and 9-(carboxymethoxy)methylguanine (CMMG) in Japanese chronic hemodialysis patients following a single oral administration of 1000 mg VACV and the influence of genetic polymorphisms of aldehyde dehydrogenase-2 (ALDH2) on their pharmacokinetics. A total of eighteen individuals genotyped as ALDH2*1/*1, ALDH2*1/*2 or ALDH2*2/*2 were enrolled in this study. Blood samples were obtained pre-dose and up to 48 hour post-dose. ACV $t_{1/2}$ was significantly affected by ALDH2 genotype and prolonged in the order of ALDH2*1/*1 (18.1 hr) < ALDH2*1/*2 (21.9 hr) < ALDH2*2/*2 (26.7 hr). ACV AUC from zero to infinity ($AUC_{0-\infty}$) increased with prolonged ACV $t_{1/2}$. ACV C_{max} was similar across the three ALDH2 genotype groups. There was no apparent relationship between ALDH2 genotype and VACV or CMMG pharmacokinetics. This is the first study to show an association between ALDH2 genetic polymorphisms and ACV elimination rate ($t_{1/2}$) in Japanese end-stage renal disease patients.

Keywords: valaciclovir; aciclovir; CMMG; end-stage renal disease (ESRD); hemodialysis; pharmacokinetics; aldehyde dehydrogenase; genetic polymorphisms

Introduction

ALDH2 is the major enzyme responsible for the elimination of acetaldehyde, produced after intake of alcohol. Over 40% of Japanese people exhibit reduced ALDH2 activity because of an ALDH2 variant allele (ALDH2*2) and are known to suffer aldehyde-induced headache, nausea and other uncomfortable symptoms even after drinking a small amount of alcohol.[1)] However this phenomenon is less common in Caucasians and Africans in whom the ALDH2*2 allele is rare.[2)] In Japanese, the three ALDH2 genotypes at frequencies of about 56% for ALDH2*1/*1, 40% for ALDH2*1/*2 and 4% for ALDH2*2/*2, results in an allele frequency of about 0.24 for ALDH2*2.[3,4)] The acetaldehyde oxidation activity of ALDH2 is high for ALDH2*1/*1, intermediate for ALDH2*1/*2, and inactive for ALDH2*2/*2.[5)]

Valaciclovir (VACV) is an oral prodrug that provides significantly enhanced aciclovir (ACV) bioavailability compared with oral ACV. VACV, like ACV, is widely used for the treatment of herpes virus infections, including herpes zoster. After oral administration, VACV is rapidly absorbed from the gastrointestinal tract and readily hydrolyzed to its active metabolite ACV that exerts potent antiviral activity against herpes simplex viruses and chickenpox/varicella zoster viruses.[6–8)] ACV is eliminated primarily by renal excretion through glomerular filtration and tubular secretion. A proportion of ACV is further converted to CMMG and 8-OH ACV.[9)] When ^{14}C-aciclovir was administered by one hour intravenous infusion, drug-related compounds excreted in urine were principally ACV (62–91%), followed by CMMG (8.5–14.1%) and 8-OH ACV (less than 0.2%).[10)]

In patients with renal failure receiving VACV, the elimination of ACV depends more upon the metabolic pathway which normally accounts for a small proportion

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**To whom correspondence should be addressed:* Katsutoshi HARA, *Clinical Pharmacology Department, Development and Medical Affairs Division, GlaxoSmithKline K.K., GSK Bldg. 6-15, Sendagaya 4-chome, Shibuya-ku, Tokyo 151-8566 Japan. Tel. +81-3-5786-5078, Fax. +81-3-5786-5219, E-mail: Katsutoshi.2.Hara@gsk.com*

of ACV elimination.[11] The degree of this dependence increases with decreasing renal function. In patients whose kidneys stop functioning completely, the metabolic pathway is the sole elimination route for ACV. ACV is thought to be converted to CMMG by alcohol dehydrogenase (ADH) and aldehyde dehydrogenase (ALDH).[9] Consequently, it is expected that the conversion rate of ACV to CMMG would be decreased in ALDH2*1/*2 and *2/*2 genotypes, thereby leading to higher plasma concentrations of ACV in *2-carrying end-stage renal disease (ESRD) subjects than non-*2-carrying ESRD patients.

In this study, we focused on the genetic polymorphisms of ALDH2 implicated in the variability of acetaldehyde degrading activity and investigated the pharmacokinetics of VACV, ACV and CMMG in Japanese chronic hemodialysis patients following single oral administration of 1000 mg VACV. The study (Study ID: 104689) was an open-label design performed with the primary objective of investigating the relationship between the ALDH2 genotype and ACV pharmacokinetics in Japanese ESRD patients.

Methods

This study was conducted at the Shitoro Clinic and Shitoro Clinic Arai (Shizuoka, Japan) between July and August 2005. This study was approved by the relevant local Institutional Review Board in Japan, and informed consent was obtained in writing from each volunteer before entry into the study. This study was conducted in accordance with the Declaration of Helsinki (South Africa, 1996) and Good Clinical Practice (GCP).

Subjects: The subjects were selected on the basis of the following criteria: 1) All subjects were at least 20 years old at the time of informed consent. 2) They were Japanese subjects with chronic renal failure undergoing hemodialysis three times a week for at least 12 weeks prior to the start of the study. 3) They gave written consent to participate in the study (including Pharmacogenomics research).

Eighteen subjects (six ALDH2*1/*1, six ALDH2*1/*2, and six ALDH2*2/*2) gave consent and were enrolled into the study. The genotype of ALDH2 of all subjects had been determined using the Invader® DNA assay in BML Inc. (Tokyo, Japan).

The sample size of 18 was chosen based on the feasibility to obtain pharmacokinetic information and not on statistical considerations.

Study schedule: The trial medication was administered to the subjects on the next day (Day 1) following their regular hemodialysis program. After the last sample collected at 48 hours post-dose (Day 3), the subjects underwent a four hour hemodialysis session. On subsequent hemodialysis days (Day 5 and 7), all participating subjects returned to the clinic.

Hemodialysis: Hemodialysis was performed for 4 hours using a double-needle access in the arteriovenous fistula and a hollow-fibre dialyser with a filter surface area of 1.5–2.1 m^2. The dialysate flow was 500 mL/min and the blood flow was 200–230 mL/min.

Dose and administration: The investigator administered two 500 mg VACV HCl tablets as a single dose of 1000 mg VACV with 100 mL of water, the subjects were not allowed food and drink (except water) for 12 hours before, and four hours after the medication was administered. The VACV HCl tablets supplied as the marketed formulations, were white to pale yellowish-white film-coated tablets, each containing 556 mg of VACV HCl (equivalent to 500 mg of VACV).

Blood and Urine sampling: Blood sampling (1 mL: plasma) for determining VACV, ACV and for its metabolite CMMG (2 mL: serum) were taken into a vacuum tube at the following intervals, just before the administration, and 1, 2, 3, 4, 6, 8, 12, 24, 44 and 48 hours after the administration of VACV. To evaluate the percentage of ACV and CMMG removed by hemodialysis, blood samples were taken before and after the four hour hemodialysis sessions on Day 3, Day 5 and Day 7. Plasma and serum were kept at −70°C or below until the analyses were performed. For calculation of creatinine clearance (CL_{cr}), blood samples were taken pre-dose, 24 and 48 hours post-dose and urine collections were performed at 0–24 and 24–48 hours post-dose. The creatinine concentrations of these samples were determined, commonly used calculation method for CL_{cr} was applied.

Determination of ACV, VACV and CMMG concentrations in plasma or serum: Plasma concentrations of ACV and VACV and serum concentrations of CMMG were determined using validated liquid chromatography mass spectrometry (LC/MS/MS) methods. The within and between coefficients of variation of the precision of these methods were less than 10.5% and the accuracies (%Dev)* were less than ±9.0%. The lower limits of quantification (LLQ) were 10 ng/mL for plasma ACV, 10 ng/mL for plasma VACV and 16 ng/mL for serum CMMG.

Pharmacokinetic analyses and percentage removed by hemodialysis: Pharmacokinetic analysis of ACV, VACV and CMMG was carried out with by standard noncompartmental methods using WinNonlin (Pharsight, CA, version 3.1). The peak concentration (C_{max}) and concentration peak time (t_{max}) were obtained directly from the observed data. The terminal rate constant (ke) was obtained by regression analysis of the terminal log-linear part of the concentration-time curve for each subject. The elimination half-life ($t_{1/2}$) was calculated as 0.693/ke. The area under the plasma concentration-

*: %Dev = [(Mean measured concentration/Nominal concentration) −1]*100

time curve (AUC_{0-12}) was calculated using the linear/log trapezoidal rule. AUC from zero to infinity ($AUC_{0-\infty}$) was calculated by $AUC_{0\text{-last}} + C_{last}/ke$. The percentages of plasma ACV and serum CMMG removed from the systemic circulation during hemodialysis were calculated from their concentrations before and after a four hour hemodialysis session.

Safety analyses: The safety profile was characterized using adverse events, change in body weight, and clinical laboratory measurements.

Statistical analyses: For the derived pharmacokinetic parameters (C_{max}, t_{max}, AUC_{0-12}, $AUC_{0-\infty}$, $t_{1/2}$), summary statistics were calculated by ALDH2 genotype and overall for subjects in the study. The pharmacokinetic parameters for plasma ACV were analyzed using a linear model incorporating ALDH2 genotype as an explanatory variable. Using this model, the significance of the ALDH2 genotype was tested and the coefficient of determination (r^2) due to genotype was estimated from the residual mean square. The PROC MIXED of SAS (SAS Institute, Cary, NC, version 8.2) was used to perform all statistical analyses. No adjustments were made for multiplicity with repeated analyses.

Results

All 18 subjects who took part in the study completed the protocol-specified observations and examinations, no protocol deviations occurred, and no subjects were withdrawn from the study.

Demographic and other baseline characteristics: **Table 1** summarizes the demographic and baseline characteristics of the 18 subjects enrolled in the study. Hemodialysis histories of these patients were at least one year or more, ten patients exceeded five years. CL_{cr} was below 2.94 mL/min in all subjects, which indicates sever renal impairment with very little residual kidney function. The demographic and baseline characteristics were similar across the three ALDH2 genotype groups.

Pharmacokinetic analyses: Mean plasma ACV, VACV and serum CMMG concentration-time profiles are displayed by ALDH2 genotype in **Figs. 1–3**. Summary

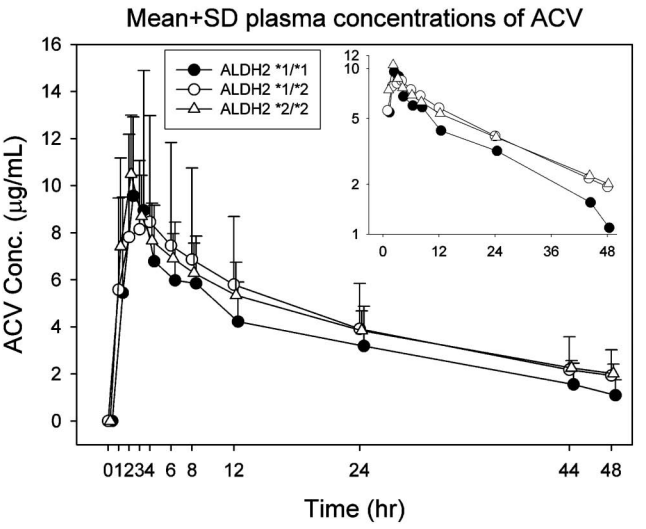


Fig. 1. Mean plasma concentrations of aciclovir (ACV) in ALDH2*1/*1 group (closed circles: $n=6$), ALDH2*1/*2 group (opened circles: $n=6$) and ALDH2*2/*2 group (opened triangles: $n=6$) following a single oral dose of 1000 mg valaciclovir. The bracketed bars indicate SD.

Table 1. Demographic and other characteristics in ALDH2 genotype group

Characteristic		ALDH2 genotype group			Total
		*1/*1 n=6	*1/*2 n=6	*2/*2 n=6	n=18
Gender	Male	4	3	4	11
	Female	2	3	2	7
Age (years)		53.5±14.8[a)]	57.5±11.0	59.7±5.8	56.9±10.8
Weight (kg)		55.7±6.8	53.7±14.1	59.2±8.5	56.2±10.0
BMI (kg/m^2)		21.0±3.5	21.0±3.2	22.5±2.3	21.5±2.9
Alcoholic beverage consumption	No	3	5	5	13
	Yes	3	1	1	5
Hemodialysis history	<1 yr	0	0	0	0
	1 yr=<, <2 yrs	0	1	1	2
	2 yrs=<, <5 yrs	3	1	2	6
	5 yrs=<, <10 yrs	2	1	2	5
	10 yrs=<	1	3	1	5
Creatinine clearance[b)] (mL/min)		0.93±0.82	0.72±0.72	1.14±1.15	0.93±0.88

a) Mean±SD, b) Calculated with urine collection

statistics for ACV, VACV and CMMG pharmacokinetic parameters are shown in **Table 2**. The results of linear model statistical analyses are shown in **Table 3**.

The means ± SD of ACV $t_{1/2}$ in the ALDH2*1/*1 group, ALDH2*1/*2 group and ALDH2*2/*2 group were 18.1 ± 4.1 hr, 21.9 ± 3.2 hr, and 26.7 ± 4.0 hr, respectively (**Table 2**). A statistically significant association of ACV $t_{1/2}$ with ALDH2 genotype was shown (linear model: $p = 0.005$) (**Table 3**). The ACV $t_{1/2}$ value was significantly longer in the ALDH2*2/*2 group than in the ALDH2*1/*1 group (*1/*1 vs. *2/*2: p = 0.001) (**Fig. 4**). The ACV $t_{1/2}$ in the ALDH2*1/*2 group and ALDH2*2/*2 group were 1.21 and 1.48 times that in the ALDH2*1/*1 group, respectively. This genetic polymorphism of ALDH2 accounted for 51% of the observed in-

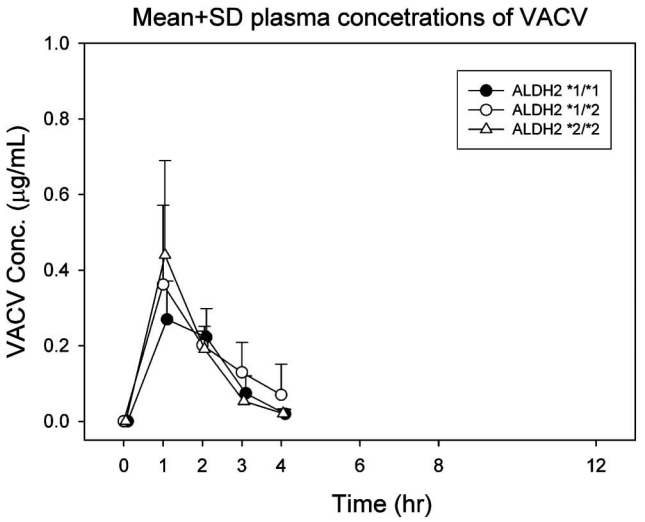


Fig. 2. Mean plasma concentrations of valaciclovir (VACV) in ALDH2*1/*1 group (closed circles: $n = 6$), ALDH2*1/*2 group (opened circles: $n = 6$) and ALDH2*2/*2 group (opened triangles: $n = 6$) following a single oral dose of 1000 mg valaciclovir. The bracketed bars indicate SD.

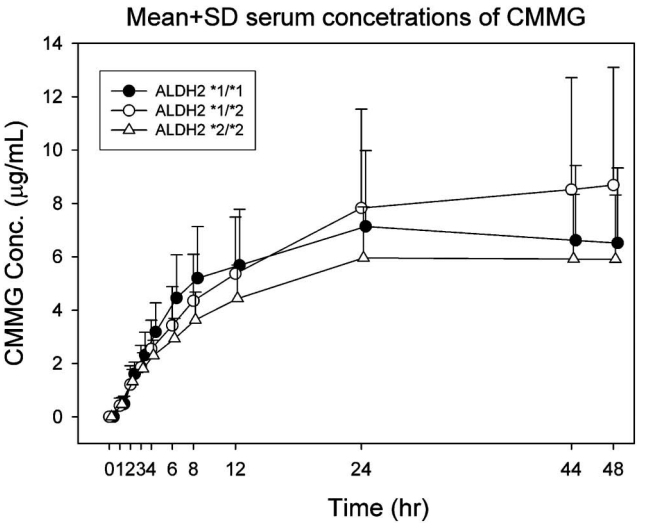


Fig. 3. Mean serum concentrations of 9-(carboxymethoxy)methylguanine (CMMG) in ALDH2*1/*1 group (closed circles: $n = 6$), ALDH2*1/*2 group (opened circles: $n = 6$) and ALDH2*2/*2 group (opened triangles: $n = 6$) following a single oral dose of 1000 mg valaciclovir. The bracketed bars indicate SD.

Table 2. The pharmacokinetic parameters of aciclovir (ACV), valaciclovir (VACV) and CMMG after oral administration of a single dose of 1000 mg VACV in ALDH2 genotype group)

Parameter	ALDH2 genotype group			Total
	*1/*1 n = 6	*1/*2 n = 6	*2/*2 n = 6	n = 18
ACV				
$t_{1/2}$ (hr)	18.1 ± 4.1[a)]	21.9 ± 3.2	26.7 ± 4.0	22.2 ± 5.0
t_{max} (hr)	2.00(2.00–3.00)[b)]	3.00(2.00–4.00)	2.00(1.00–3.00)	2.00(1.00–4.00)
C_{max} (μg/mL)	10.96 ± 5.52	9.80 ± 4.66	11.05 ± 2.70	10.60 ± 4.22
$AUC_{0-\infty}$ (μg·hr/mL)	197.62 ± 93.49	270.43 ± 146.94	280.23 ± 48.22	249.43 ± 105.09
AUC_{0-12} (μg·hr/mL)	71.55 ± 26.27	81.13 ± 40.61	81.23 ± 16.54	77.97 ± 28.11
VACV				
$t_{1/2}$ (hr)	0.52 ± 0.12	0.77 ± 0.21	0.63 ± 0.13	0.64 ± 0.18
t_{max} (hr)	1.00(1.00–2.00)	1.00(1.00–2.00)	1.00(1.00–1.00)	1.00(1.00–2.00)
C_{max} (μg/mL)	0.31 ± 0.08	0.37 ± 0.20	0.44 ± 0.25	0.37 ± 0.19
$AUC_{0-\infty}$ (μg·hr/mL)	0.564 ± 0.088	0.778 ± 0.269	0.674 ± 0.275	0.672 ± 0.232
CMMG[c)]				
t_{max} (hr)	24.0(8.0–48.0)	48.0(24.0–48.0)	24.0(24.0–48.0)	34.0(8.00–48.0)
C_{max} (μg/mL)	7.44 ± 2.79	8.75 ± 4.31	6.22 ± 2.29	7.47 ± 3.23

a) Mean ± SD, b) Median (range), c) 9-(carboxymethoxy)methylguanine

Table 3. Results of analysis of variance and corresponding r^2 values of study dependent variables due to ALDH2 genotype

Dependent variable	F	p value[a)]	r^{2} [b)]
ACV			
$t_{1/2}$	7.8	0.005**	0.510
C_{max}	0.15	0.864	0.019
$AUC_{0-\infty}$	1.12	0.350	0.130
AUC_{0-12}	0.21	0.811	0.028

a) statistical significant of ALDH2 genotype, b) the coefficient of determination due to ALDH2 genotype
**: significantly different ($p<0.01$)

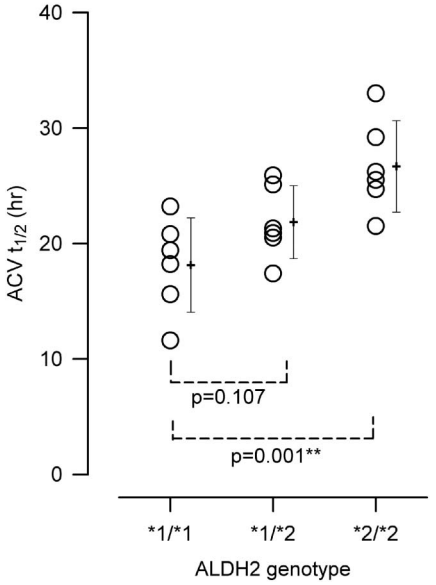


Fig. 4. Effect of ALDH2 genotype on aciclovir (ACV) $t_{1/2}$
The three groups represent ALDH2*1/*1 group, ALDH2*1/*2 group and ALDH2*2/*2 groups. Contrast test were used to compare the ALDH2*1/*1. **Indicates a significant difference ($p<0.01$). Bars Mean±SD.

terindividual variability in ACV $t_{1/2}$ and for 13% in ACV $AUC_{0-\infty}$ (**Table 3**). ACV C_{max}, AUC_{0-12} and $AUC_{0-\infty}$ were not significantly associated with ALDH2 genotype although ACV $AUC_{0-\infty}$ was numerically greater in both the ALDH2*1/*2 and ALDH2*2/*2 groups than ALDH2*1/*1 group.

There was no apparent relationship between ALDH2 genotype and VACV or CMMG pharmacokinetics (data not shown).

Percentage of removed by hemodialysis: A four hour hemodialysis session on Day 3 removed 69.4± 9.1% of ACV in plasma and 77.6±4.7% of CMMG in serum. The percentages of ACV and CMMG removed on Day 5 and 7 were similar to the value on Day 3 (data not shown).

Safety analyses: A total of two subjects (11%, 2/18) had two adverse events: one with thermal burn and the other with subcutaneous haemorrhage. Both subjects were in the ALDH2*1/*1 group. The two events were mild in severity and assessed as not related to the study drug. No clinically significant changes in laboratory values or body weight occurred during the study, nor were there any adverse events related to clinical laboratory or body weight.

Discussion

The present study demonstrated a statistical association between ALDH2 genetic polymorphisms and ACV elimination rate ($t_{1/2}$) in Japanese ESRD patients.

ACV, the active metabolite of VACV, is eliminated primarily by renal excretion.[10] In patients whose kidneys stop functioning completely, the metabolic pathway is the sole elimination route for ACV. ACV is thought to be converted to CMMG by ADH and ALDH.[9] In addition, in Japanese, it is well established that the ALDH2 genotypes *2/*2 and *1/*2 are associated with lower enzymatic activity levels.[3,4]

This study included a total of 18 subjects, six for each ALDH2 genetic polymorphism group. The mean values of ACV $t_{1/2}$ in the ALDH2 *1/*1, ALDH2 *1/*2 and ALDH2 *2/*2 groups were 18.1 hr, 21.9 hr, and 26.7 hr, respectively. Hence ACV $t_{1/2}$ was significantly prolonged in the order of decreasing ALDH2 activity (**Fig. 4**). The mean $AUC_{0-\infty}$ also increased with prolonged $t_{1/2}$ (**Table 2**) although the differences between the lower activity and wild type genotypes did not reach statistical significance. There were no clear relationships between ALDH2 genotype and C_{max} or AUC_{0-12} for ACV (**Table 2**). The present study clearly shows that the ALDH2 genetic polymorphisms influence ACV $t_{1/2}$, the prolongation of which also influences ACV $AUC_{0-\infty}$.

The results of linear model analysis showed that genetic polymorphism of ALDH2 accounted for 51% of the interindividual variability in ACV $t_{1/2}$ among the 18 subjects (**Table 3**). Because the terminal $t_{1/2}$ reflects variability in elimination only, the remaining variability of $t_{1/2}$ may potentially be explained physiologically by the residual renal reserve and the level of hepatic function, and genetically by polymorphism of ADH2, ALDH1 and/or ALDH2.[3,4,12,13]

The contribution of the genetic polymorphism of ALDH2 to $t_{1/2}$ was found to be statistically significant, but a significant association was not seen for $AUC_{0-\infty}$, which may be affected by various other factors in addition to the ALDH2 genotype, including distribution volume (potentially related to body weight) and interindividual variability in drug absorption. Furthermore, the small sample size and/or single-dose design may have precluded detection of a difference in AUC based on genotype.

A pharmacokinetic study of VACV has also been conducted in non-Japanese renally impaired patients under-

going hemodialysis. This was a study of oral VACV in nine subjects (mean body weight: 74.9 kg) (unpublished observation, GlaxoSmithKline Clinical Trial Register: http:/ctr.gsk.co.uk/Summary/valaciclovir/studylist.asp, Study ID: P66–09 (123–009)). For the overall population in the current study (n = 18, mean body weight: 56.2 kg), the mean $AUC_{0\text{-}\infty}$ for ACV administered orally as 1000 mg VACV in Japanese hemodialysis patients (249 ± 105 μg·hr/mL) was approximately two times greater than that in the non-Japanese hemodialysis patients (127 ± 27 μg·hr/mL). Similarly ACV $t_{1/2}$ in the Japanese subjects (22.2 ± 5.0 hours) was longer than in non-Japanese hemodialysis patients (14.1 ± 2.6 hours).

It is likely that this difference between Japanese and non-Japanese patients can be related in part to the existence of ALDH2*2 alleles in Japanese ESRD patients but not the non-Japanese. In order to compare appropriate Japanese and non-Japanese hemodialysis patient groups, ACV PK parameters from the Japanese ALDH2*1/*1 genotype group were used. This genotype group had an $AUC_{0\text{-}\infty}$ of 198 ± 93 μg·hr/mL and $t_{1/2}$ of 18.1 ± 4.1 hr (**Table 2**). These values are closer to although still greater than those from the non-Japanese patients. The reasons for these remaining differences are not clear, but may relate in part to the small sample sizes in each study and the difference in body weight between Japanese study (ALDH2*1/*1: 55.7 kg) and non-Japanese (74.9 kg). The results are however consistent with the observation that in patients with end-stage renal disease, the presence of ALDH2*2 alleles prolongs ACV $t_{1/2}$ and increases $AUC_{0\text{-}\infty}$.

Plasma VACV concentrations were low compared to the ACV concentrations, and fell below the limit of quantification six hours after administration (**Fig. 2**). Furthermore, there was no apparent relationship between ALDH2 genotype and VACV pharmacokinetics. These findings are consistent with the previous results that VACV was rapidly hydrolyzed to ACV by mainly VACVase (Biphenyl hydrolase-like protein).[14]

In the early-phase of the CMMG time-course, the influence of ALDH2 genetic polymorphism was seen in the rate of production of CMMG. The slope of the mean CMMG-time curves for the three genotypes (**Fig. 3**) increases in the order ALDH2*2/*2 < ALDH2*1/*2 < ALDH2*1/*1. However, in this single dose study, this relationship is not apparent later in the time course as CMMG concentrations are mainly dependent on the individual absorption and/or distribution volume characteristics of the patients rather than ALDH2 genotype. In patients whose kidneys stop functioning completely, it is thought that all ACV is eliminated through the metabolic pathway, and the resulting metabolite accumulates in the body. In the present study, the accumulation of CMMG the main metabolite of ACV, was shown (**Fig. 3**). In our study, the standard four hour hemodialysis session removed about 70% of ACV and 80% of CMMG. Similar results have been reported previously, about 57% of ACV and 64% CMMG were removed by three to four hour hemodialysis sessions in non-Japanese.[15]

In conclusion, the present study showed a significant association between ALDH2 genetic polymorphisms and ACV elimination rate ($t_{1/2}$) in Japanese ESRD patients.

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